



EXPERIMENT NO 5

NO-LOAD AND BLOCKED-ROTOR TESTS ON SINGLE PHASE INDUCTION MOTOR

AIM

- To conduct the no load and blocked rotor tests on single phase induction motor
- To find the equivalent circuit parameters
- To predetermine its performance at rated speed.

APPARATUS

Sl.No.	Name of the apparatus	Type	Range	Quantity
1.	Voltmeter			
2.				
3.				
4.	Ammeter			
5.				
6.	Wattmeter			
7.	Rheostat			

PRINCIPLE

Since a single phase induction motor does not have a starting torque, it needs special methods of starting. The stator is provided with two windings, called main and auxiliary windings, whose axes are space displaced by 90 electrical degrees. The rotor is of squirrel cage construction. The auxiliary winding is excited by a current which is out of phase with the current in the main winding, both current derived from the same supply mains. The auxiliary winding is disconnected by a centrifugal switch after the motor has achieved about 75% speed. Depending upon the starting methods, single phase induction motors are classified into i) split phase motor, ii) capacitor start motor, iii) Capacitor start and run motor, iv) shaded pole motor and v) repulsion start induction run motor. In all these methods, it will act as a two-phase motor at the time of starting.

Here, no load and blocked rotor tests on capacitor start induction motor are done. In a capacitor start motor, the main (or running) and auxiliary (or starting) windings are space displaced by 90°. The time displacement between the currents in the main and auxiliary winding is achieved by connecting a capacitor in series with auxiliary winding. By using a capacitor of proper value, the current I_a in the auxiliary winding can be made to lead the current I_m in the main winding by 90° at standstill. Thus the motor develops a starting torque. Both the capacitor and auxiliary winding are